

FitSathi

Har Student, Har Jagah, Har Budget

Problem ID:

26196

Title:

Student Innovation — Ideas that can boost fitness activities and assist in keeping fit

Organization:

AICTE

Department:

AICTE, MIC-Student Innovation

Category:

Software

Theme:

Fitness & Sports

Complete Master Blueprint & Problem-Solution Architecture

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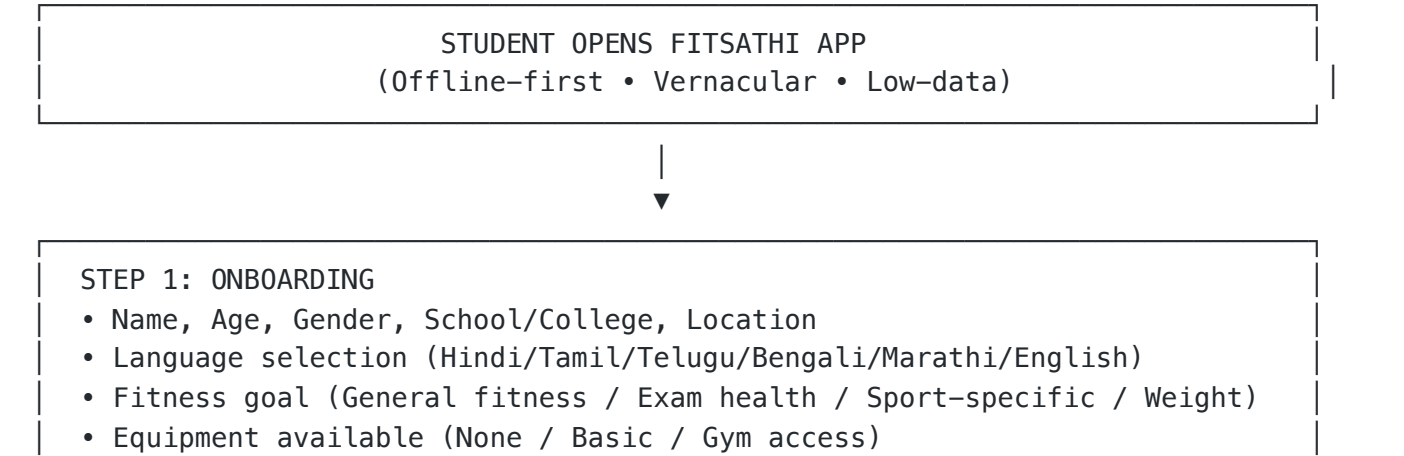
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1. Master Flow: From Registration to Nation-Building



- Parent linking (optional)



STEP 2: TALENT ID CREATION

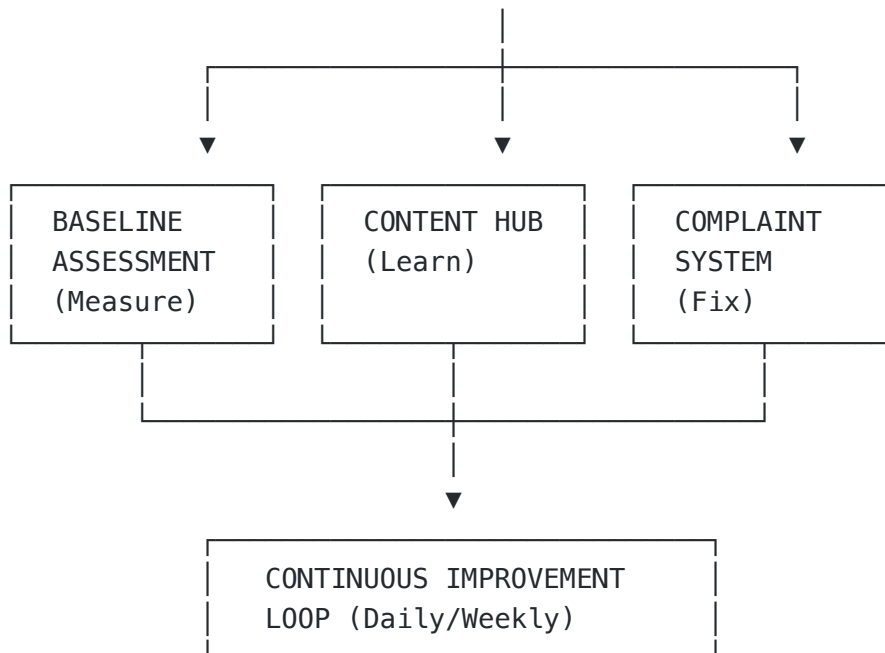
UNIQUE FITNESS PASSPORT ID generated

Format: FS-{State}-{District}-{School}-{Year}-{RollNo}

Example: FS-MH-PUN-SCH042-2025-1087

This ID becomes the single source of truth for:

- ✓ Fitness assessments
- ✓ Improvement tracking
- ✓ Talent discovery
- ✓ Complaint history
- ✓ Content progress
- ✓ Buddy matching
- ✓ School Sports Score
- ✓ Parent dashboard
- ✓ Scholarship exports



2. Pillar 1: Talent ID — Complete Flow

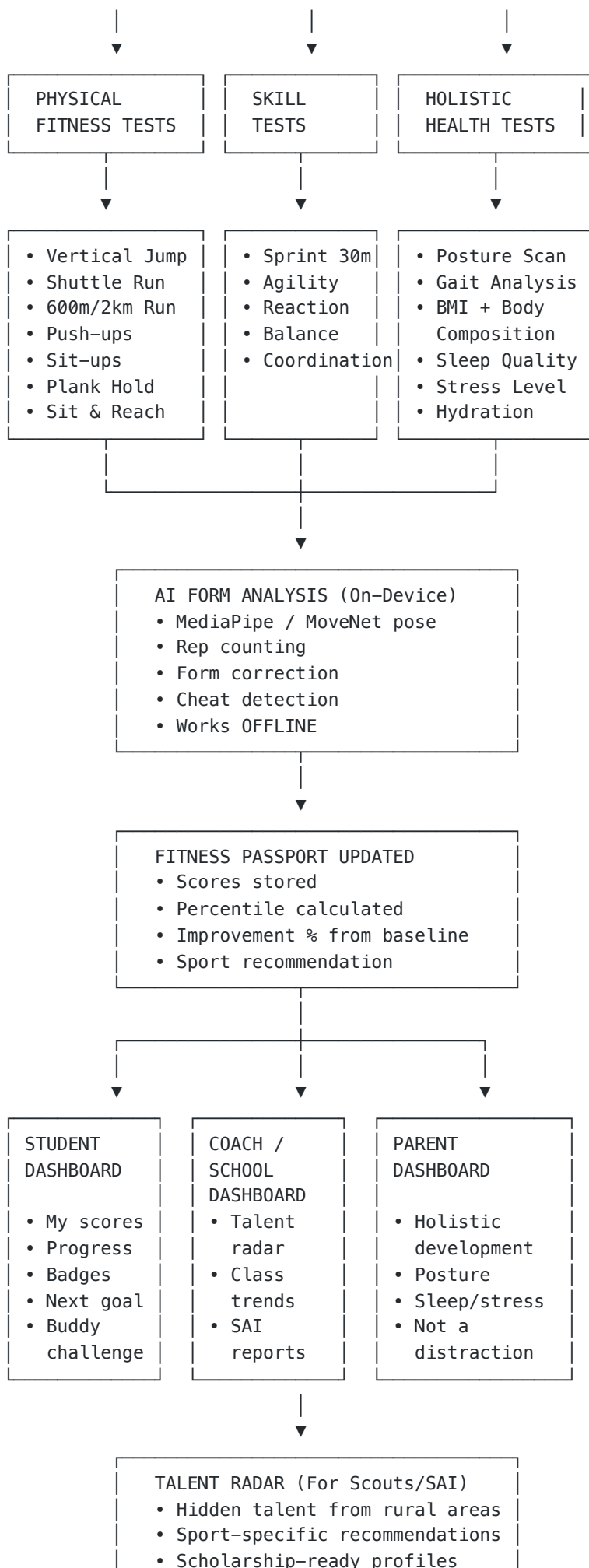
TALENT ID SYSTEM FLOW

STUDENT



TALENT ID CREATED
(Fitness Passport)





- Khelo India integration

Problems Solved by Talent ID

#	Problem	How Talent ID Solves It
1	Poor school sports system	Standardized testing + tracking for every student
2	No professional coaching	AI form coach gives feedback without a coach
3	No early talent development	Baseline at young age → long-term tracking
4	Embarrassment vs trained players	Improvement-based ranking, not absolute comparison
5	Lack of real-time guidance	On-device AI corrects form instantly
6	No progress tracking	Fitness Passport stores every result
7	Lack of personalized goals	AI recommends next target based on data
8	No healthy competition	Improvement Leaderboard (fair)
9	No recognition	Badges, certificates, SAI reports
10	Economic/rural gap	Works offline, no equipment needed, vernacular
11	Parental mindset	Parent dashboard shows holistic growth
12	Exam posture epidemic	Posture scan + gait analysis
13	Dysmorphia trap	Realistic goals, improvement not appearance
14	Female safety	Home-based tests, no outdoor needed

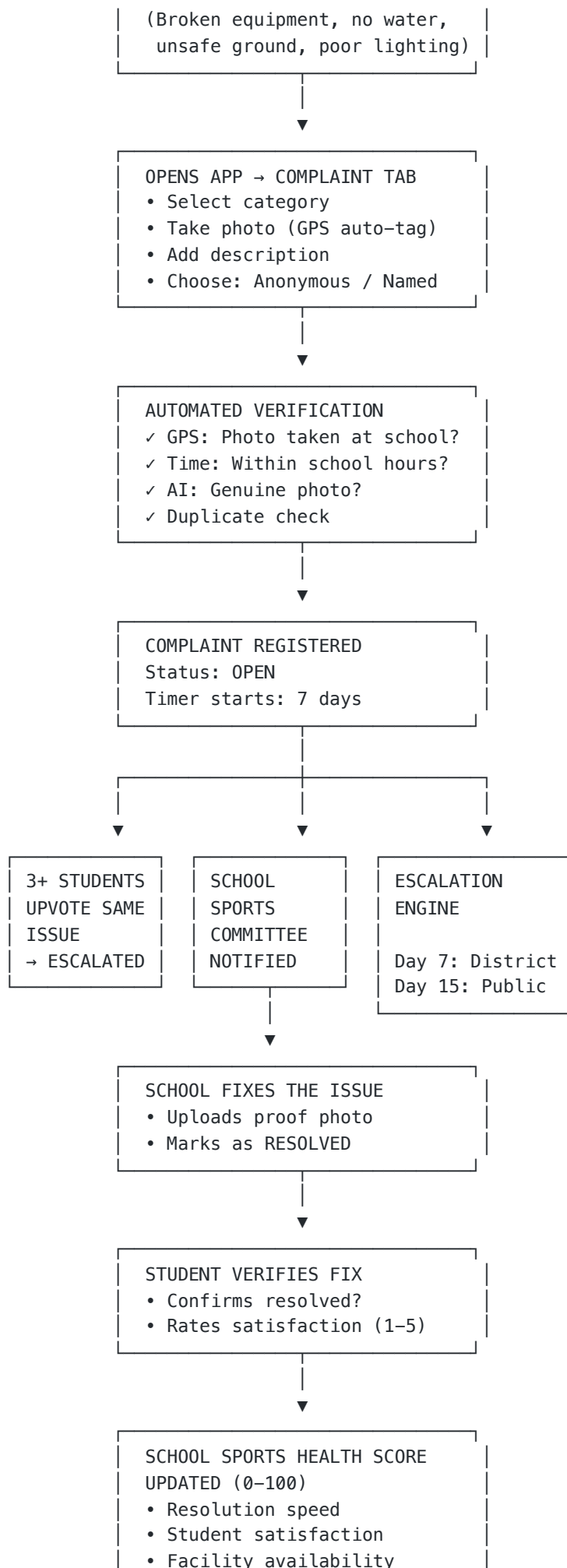
3. Pillar 2: Complaint & Inspection System — Complete Flow

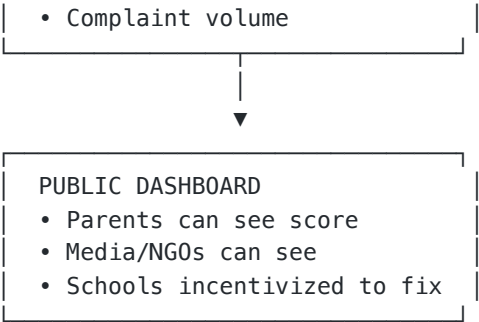
STUDENT-ONLY COMPLAINT & INSPECTION SYSTEM

STUDENT (Verified via Talent ID)



STUDENT SPOTS A PROBLEM



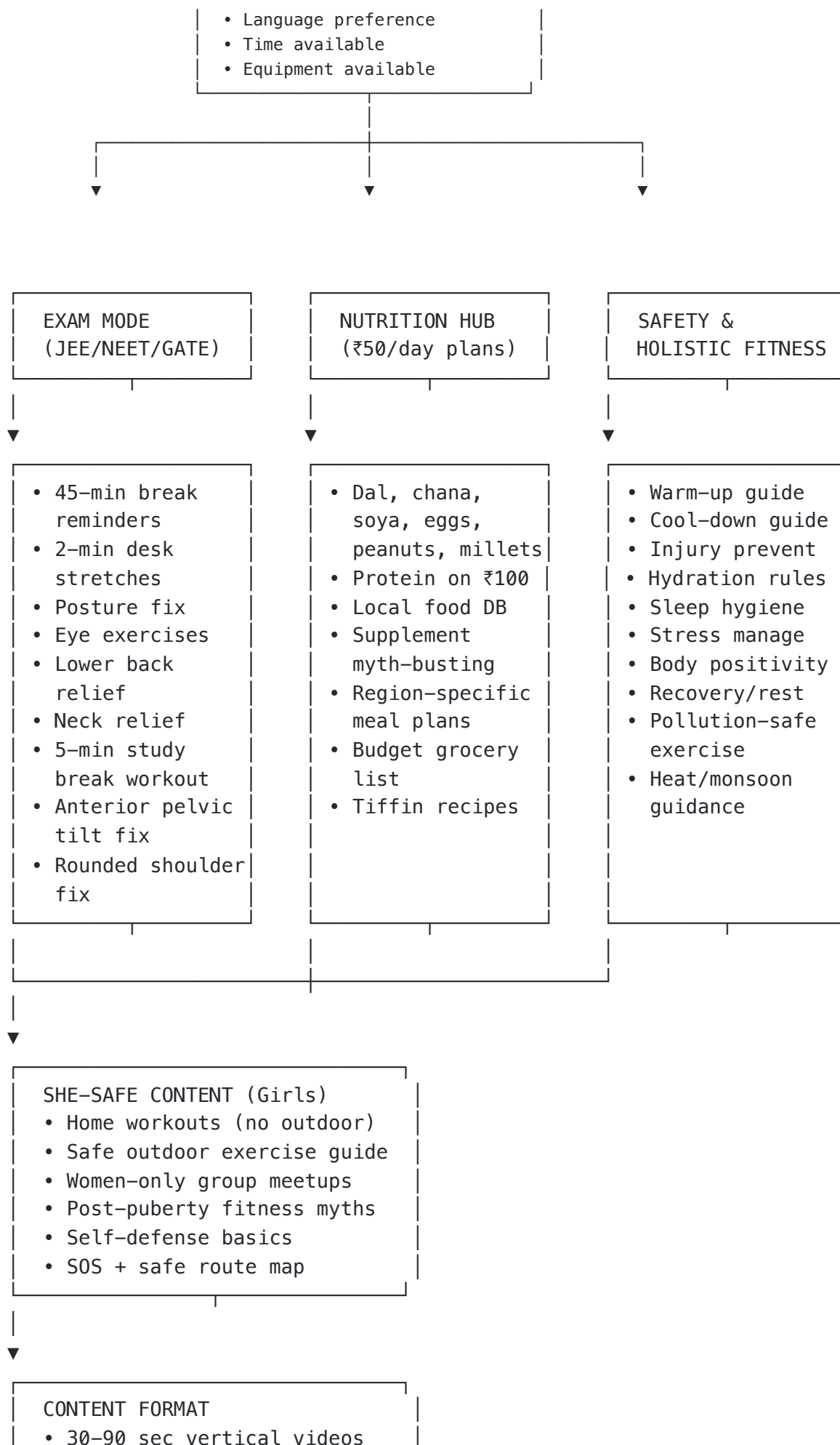


Complaint Categories & Problems Solved

Category	Problem It Addresses
Broken Equipment	Lack of accessible sports equipment
Unsafe Ground	Safety issues, overcrowding
No Drinking Water	Lack of hydration facilities
Poor Lighting	Female safety barrier
No Toilet/Changing Room	Female dropout post-puberty
Overcrowding	Shared grounds (walkers vs players)
No Running Track	Lack of proper infrastructure
No Coach	Poor school sports system
No Sports Period	Academic pressure, parental mindset
Air Pollution Zone	Environmental problems

4. Pillar 3: Content Hub — Complete Flow





- Audio guides (5–10 min)
- Infographics
- Live Q&A weekly
- Offline download
- Multi-language toggle



COMPLETION TRACKED

- Watched → Practiced → Assessed → Improved
- Feeds back into Talent ID

5. Safe & Fit Map + Green Window — Complete Flow

SAFE & FIT MAP + GREEN WINDOW FLOW

STUDENT WANTS TO EXERCISE OUTDOORS



APP CHECKS:

- AQI (CPCB API)
- Weather (temp, humidity)
- Heat index
- Rain forecast
- UV index



GREEN WINDOW
Safe to exercise outdoors

YELLOW WINDOW
Caution (mask, shorter)

RED WINDOW
Unsafe (stay indoor)



SHOW MAP:

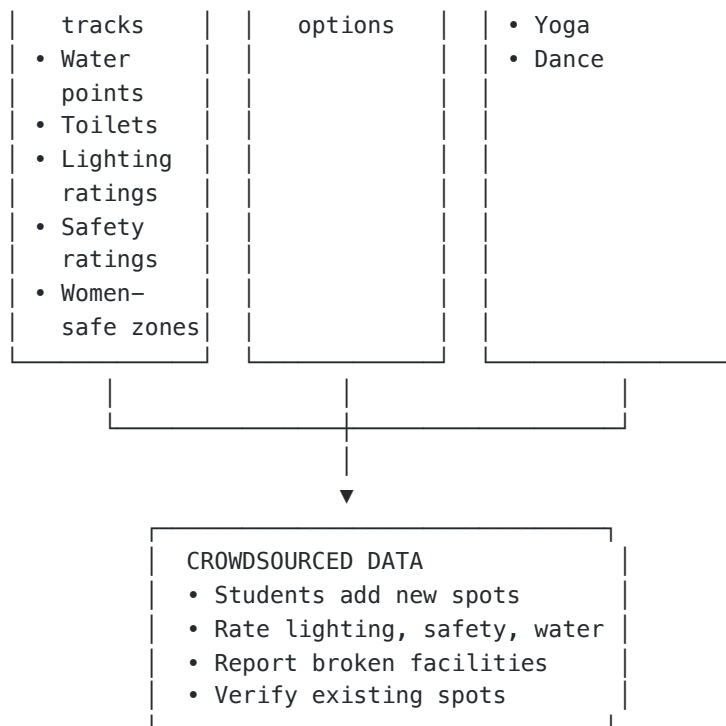
- Free grounds
- Open gyms
- Running

SHOW MAP:

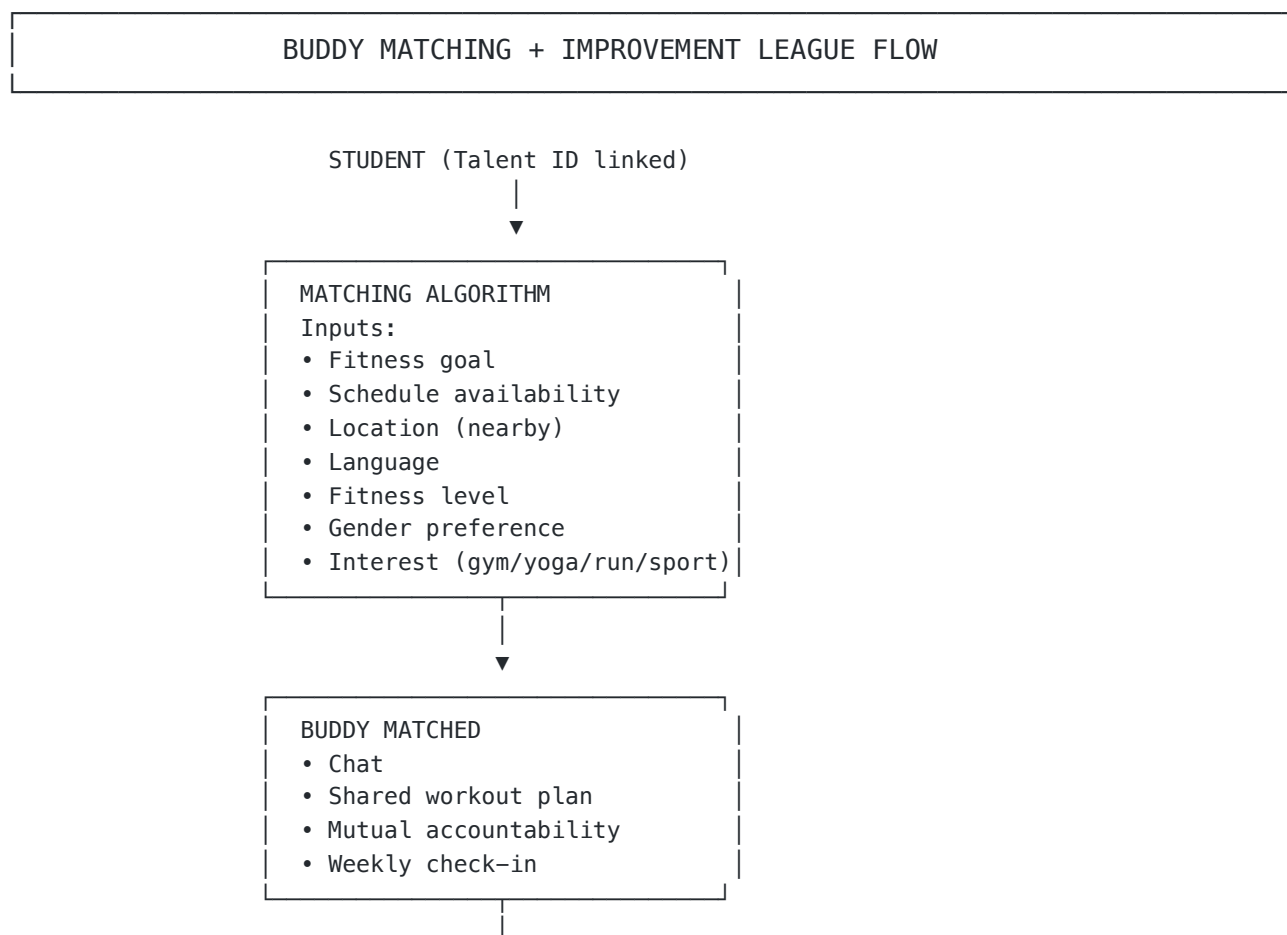
- Indoor options
- Shorter outdoor

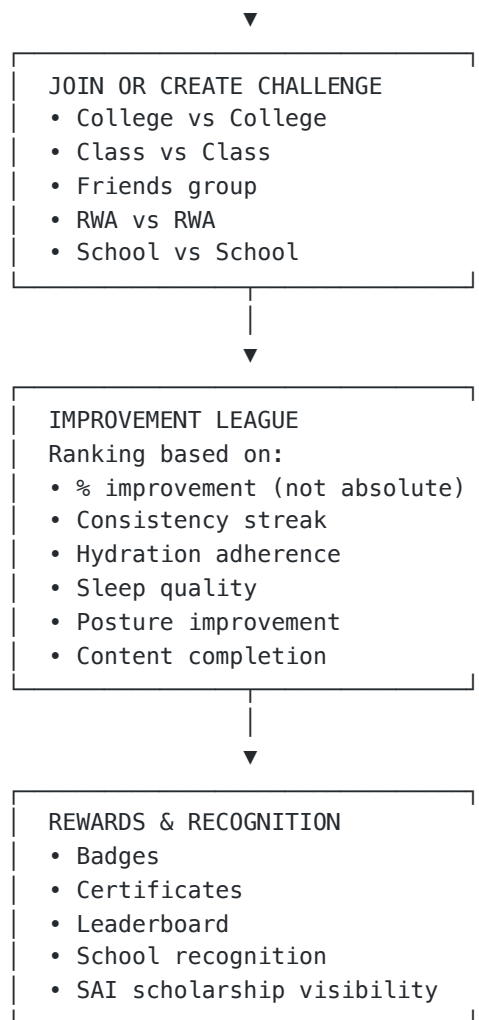
AUTO-SWITCH TO INDOOR WORKOUT

- Home bodyweight
- Exam Mode stretches



6. Buddy System + Improvement League — Complete Flow





7. Complete Problem → Solution Mapping (All 54 Problems)

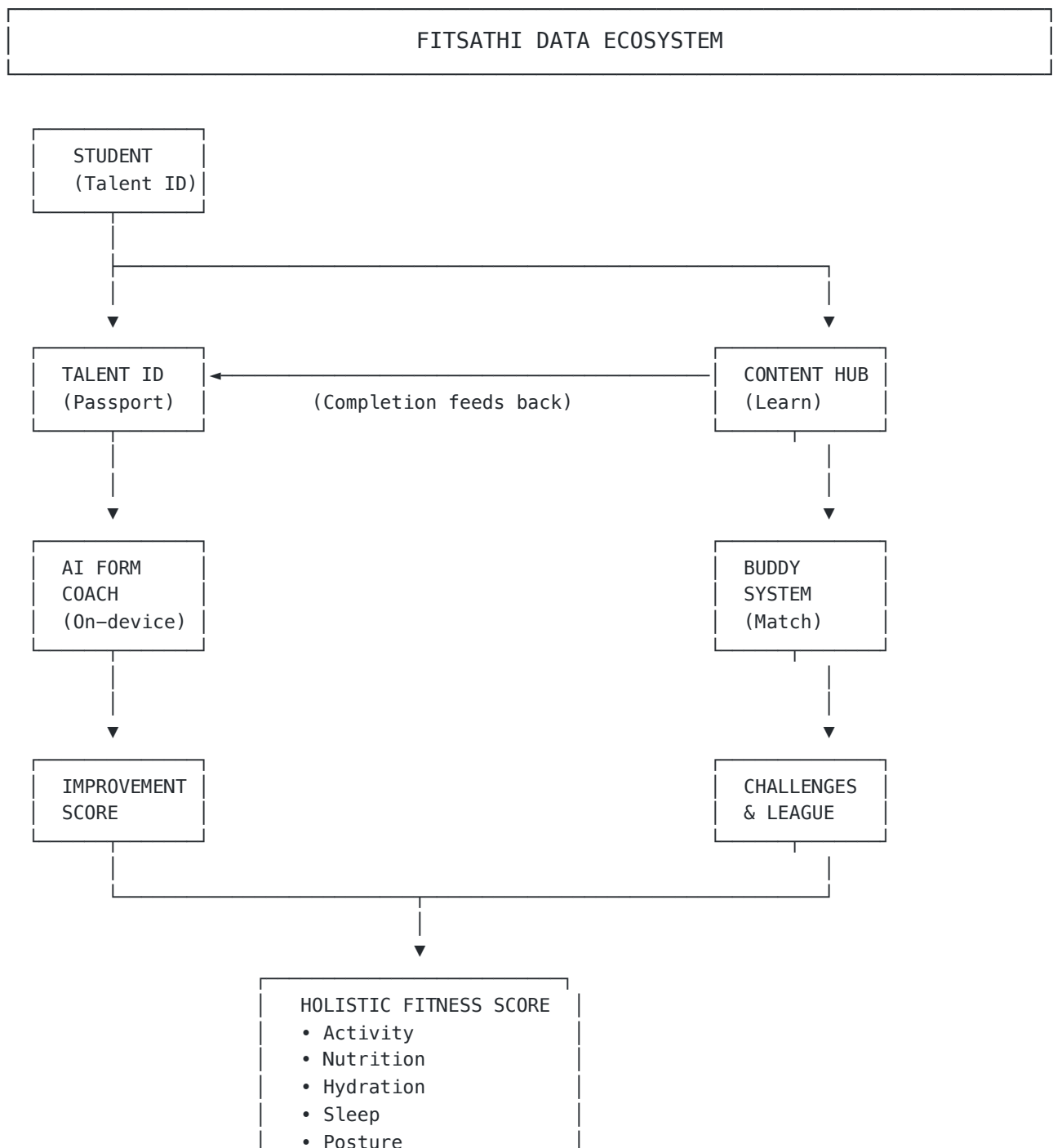
#	Problem	Solution(s)
1	Poor school sports system	Talent ID + School Dashboard + Complaint System
2	Limited sports periods	Exam Mode + 5-min study break workouts
3	Lack of professional coaching	AI Form Coach (on-device)
4	No early talent development	Talent ID baseline + long-term tracking
5	Embarrassment vs trained players	Improvement-based ranking (not absolute)
6	Lack of free infrastructure	Safe & Fit Map (crowdsourced)
7	Public grounds used for walking	Complaint System + Safe & Fit Map

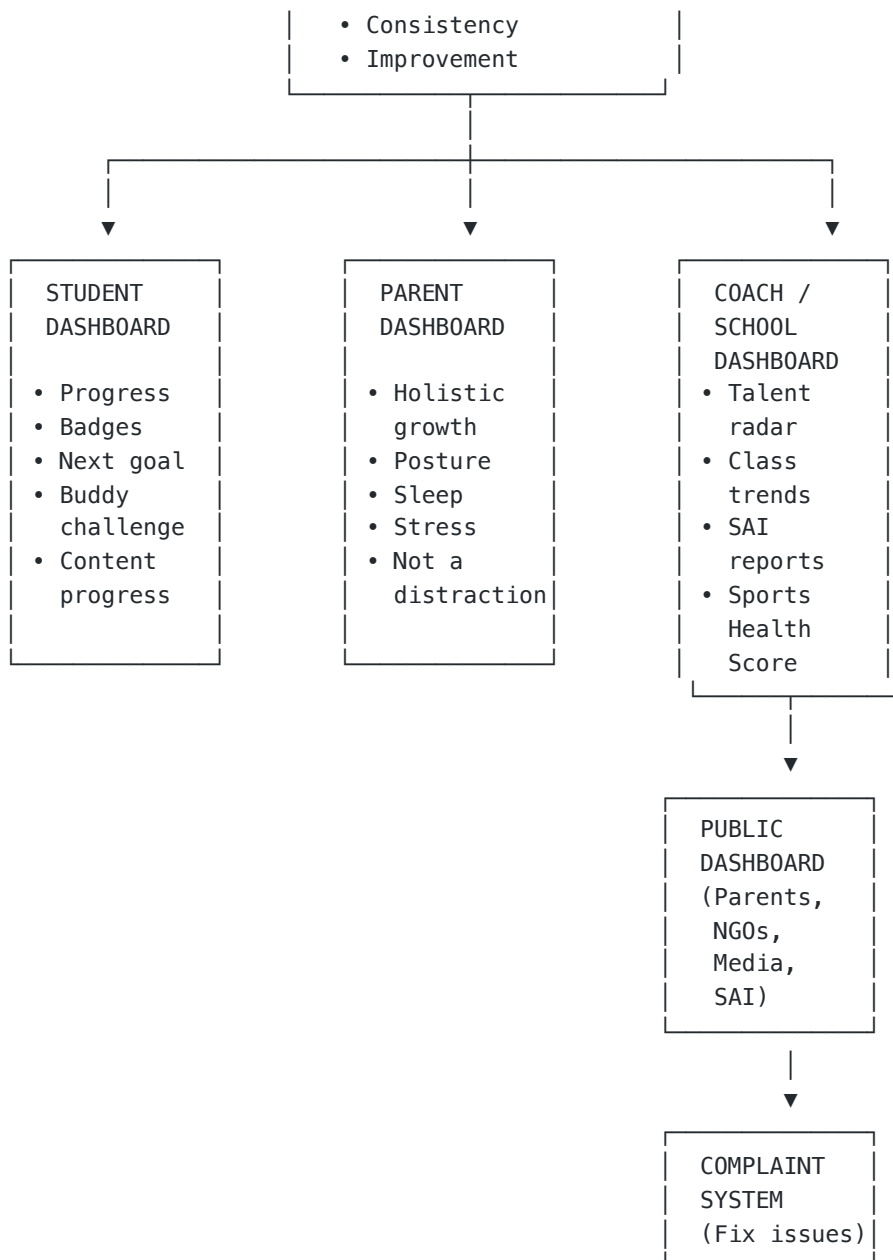
#	Problem	Solution(s)
8	Overcrowding & safety issues	Complaint System + Crowdsourced ratings
9	No running tracks	Safe & Fit Map + Complaint System
10	Lack of accessible equipment	No-equipment workouts + Complaint System
11	Fitness is expensive	₹50/day nutrition + no-gym plans + free AI coach
12	Healthy food unaffordable	Indian nutrition DB + budget meal plans
13	Lack of fitness companionship	Buddy Matching System
14	Boring to exercise alone	Buddy System + Improvement League + Challenges
15	Lose motivation without friends	Buddy + Leaderboard + Badges
16	Lack of real-time guidance	AI Form Coach (on-device, offline)
17	Don't know correct intensity	Adaptive plans + AI recommendations
18	No feedback on improvement	Fitness Passport + Improvement Score
19	No personalized goals	AI-generated weekly goals from Talent ID data
20	Lack of motivation & consistency	Streaks + Badges + Challenges + Rewards
21	No challenges/rewards	Improvement League + School Sports Score
22	Air pollution & environment	Green Window (AQI/weather) + indoor alternatives
23	Extreme heat/humidity/rain	Green Window + indoor workouts
24	Don't know safe time/place	Green Window + Safe & Fit Map
25	Lack of hydration facilities	Safe & Fit Map (water points) + Hydration reminders
26	No hydration awareness	Content Hub (hydration module)
27	No affordable nutrition guidance	₹50/day plans + local food DB + tiffin recipes
28	Healthy food perceived expensive	Budget grocery lists + myth-busting
29	No personalized diet guidance	AI meal plans based on region, budget, goal

#	Problem	Solution(s)
30	Lack of safety awareness	Content Hub (warm-up, cool-down, injury prevention)
31	Incorrect technique → injuries	AI Form Coach + Content Hub
32	Inappropriate intensity	Adaptive plans + AI recommendations
33	No recovery/rest awareness	Content Hub (sleep, recovery, rest days)
34	Economic & rural accessibility	Offline-first + vernacular + no-equipment + free
35	Fewer rural sports facilities	Safe & Fit Map + Complaint System + Talent ID
36	Parental mindset	Parent Dashboard (holistic development)
37	Sports seen as distraction	Parent Dashboard + Academic-Fitness balance reports
38	Fitness only viewed as exercise	Holistic Fitness Score (activity+nutrition+sleep)
39	No healthy competition	Improvement League (fair, improvement-based)
40	No platform to compare	Buddy System + Leaderboard + Challenges
41	Competition favors already-fit	Improvement Score (rewards % growth, not baseline)
42	Exam posture epidemic	Exam Mode + Posture Scan + Gait Analysis
43	Anterior pelvic tilt	Exam Mode stretches + AI posture correction
44	Rounded shoulders	Exam Mode + Wall angel tutorials
45	Tight hip flexors	Exam Mode + Seated hip flexor stretches
46	Chronic lower back pain	Exam Mode + Cat-cow + Child's pose
47	Western influencer trap	Realistic Indian goals + no extreme transformations
48	Dysmorphia from social media	Body positivity content + improvement-based ranking
49	Overwhelmed by 6-day PPL	Simple, sustainable plans (3-4 days, 20-30 min)
50	Discouraged by slow results	Improvement tracking + small wins celebration
51	Safety barrier for female	SHE-SAFE Mode + Safe & Fit Map + women-only groups

#	Problem	Solution(s)
52	Poorly lit streets/grounds	Complaint System + Safe & Fit Map (lighting ratings)
53	Female dropout post-puberty	SHE-SAFE + home workouts + women-only challenges
54	No safe dedicated spaces	Safe & Fit Map + Complaint System + SHE-SAFE

8. Complete Data Flow Diagram





9. SIH Demo Flow (Three Personas)

LIVE DEMO SCRIPT

PERSONA 1: RAVI (Rural, No Gym, Hindi)

0:00 Opens app offline → Hindi interface

0:30 Watches Exam Mode stretch video

1:00 Takes Vertical Jump test → AI analyzes form → Talent ID updated

1:30 Gets matched with buddy nearby → joins challenge

2:00 Sees "Top 20% in your age group" → Improvement Score rises
2:30 Parent dashboard shows holistic growth

PERSONA 2: PRIYA (Female, Safety Concern, Tamil)

2:30 Opens SHE-SAFE map → finds well-lit park with water point
3:00 Reports broken light anonymously → 5 students upvote
3:30 School fixes in 3 days → Sports Health Score improves 45→62
4:00 Joins women-only walking group → Buddy system
4:30 Watches home workout video (no outdoor needed)

PERSONA 3: ARJUN (JEE Aspirant, English/Hindi)

4:30 Studies 12 hours → Exam Mode reminds him to stretch
5:00 AI corrects posture → Parent dashboard shows improved posture
5:30 Compares improvement with classmates on Improvement Leaderboard
6:00 Coach dashboard → Talent Radar identifies 3 students for athletics
6:30 Exports Fitness Passport reports for SAI scholarship

CLOSING PITCH:

"We are not building a fitness app. We are building the digital infrastructure for student sports in India. A Fitness Passport that discovers talent from every village. A complaint system that holds schools accountable. A content hub that teaches fitness in every language. This is how India becomes a sporting nation – one student at a time."

10. Tech Stack Summary

Layer	Technology	Purpose
Mobile App	Flutter / React Native	Cross-platform, offline-first
On-Device AI	TensorFlow Lite, MediaPipe, MoveNet	Pose estimation, form analysis
Backend	Node.js + FastAPI	Scalable, real-time
Database	Firebase Firestore + PostgreSQL/PostGIS	Offline sync + geospatial
Maps	Mapbox / Google Maps SDK	Safe & Fit Map, complaints
AQI/Weather	CPCB API, OpenWeather	Green Window

Layer	Technology	Purpose
Video Hosting	Firebase Storage + Cloudflare CDN	Content Hub
Auth	Firebase Auth (anonymous option)	Student-only complaints
Admin Dashboard	React.js + Recharts	School Sports Health Score
CMS	Strapi / Sanity	Content management

11. Why FitSathi Wins

Criteria	How FitSathi Wins
Innovation	On-device AI + Student-only complaint + Fitness Passport + Vernacular content — no existing app combines all
Feasibility	All tech available, open-source. MVP in 36 hours
Social Impact	Rural access, female safety, exam health, affordability, parental mindset
Scalability	Metro to village. Offline-first. Multi-language
Alignment	Directly addresses AICTE Student Innovation in Fitness & Sports (Software)
Demo-ability	3 personas, live AI, anonymous complaint, improvement leaderboard

12. Development Roadmap

Phase	Duration	Deliverable
Phase 1: Core MVP	Week 1–2	Login, Fitness Passport, 3 SAI tests with AI form check
Phase 2: Complaint System	Week 3	Anonymous reporting, photo + GPS, admin dashboard
Phase 3: Content Hub	Week 4	10 videos in 2 languages, offline download
Phase 4: Integration	Week 5	Connect all three pillars, leaderboard, school score
Phase 5: Polish	Week 6	UI/UX, demo script, presentation

13. Final Pitch

We are not building a fitness app. We are building the **digital infrastructure for student sports in India**. A Fitness Passport that discovers talent from every village. A complaint system that holds schools accountable. A content hub that teaches fitness in every language. This is how India becomes a sporting nation — one student at a time.

Note: All flowcharts use monospace box-drawing characters. For best PDF output, use **Ctrl+P → Save as PDF** in Chrome or Edge, enable "**Background graphics**", and set margins to **Default**.